CAUSAL INFERENCE CASE STUDY SERIES — CHAPTER 17

Causal Survival Analysis for Telco Churn

Time-to-Event Estimands, Hazard Ratio Selection Bias, IP Weighting & Parametric G-Formula

1. Business Use Case & Time-to-Event Problem Formulation

In telecommunications customer retention, estimating the impact of a baseline loyalty intervention (treatment A = 1 for a 12-month contract discount vs A = 0 for standard billing) requires evaluating subscriber retention over extended horizons (e.g., 120 months). Churn is a classic **survival outcome** (time-to-event T). Because commercial tracking periods end at a specific calendar cutoff, many subscribers remain active at the end of follow-up ($K = 120$ months). Their exact churn time is unknown—a phenomenon known as **administrative censoring** ($C = 0$).

Survival Estimand	Mathematical Notation	Operational Meaning in Telco Retention
Survival Probability S(k)	P(T > k) = P(D_k = 0)	Proportion of subscribers retained active past month k.
Cumulative Risk R(k)	1 - S(k) = P(D_k = 1)	Cumulative proportion of subscribers who churned by month k.
Discrete-Time Hazard h(k)	P(D_k = 1 | D_{k-1} = 0)	Conditional risk of churning in month k among those still active.
Counterfactual Survival S^a(k)	P(T^a > k) = P(D_k^a = 0)	Retention curve if all subscribers had received intervention a.

2. The Hazards of Hazard Ratios & Selection Bias

In industrial survival analysis, reporting a single **Hazard Ratio (HR)** from a Cox Proportional Hazards model is common practice. However, Chapter 17 of What If (Hernán & Robins) demonstrates two fundamental pitfalls of hazard ratios:

- **Non-Collapsibility & Time-Variation:** Hazards vary over time ($h(k)$), making a single summary HR a weighted average that can equal 1 even when true survival curves diverge.
- **Built-in Selection Bias:** The hazard ratio at month k conditions on subscribers who survived through month $k-1$ ($D_{k-1} = 0$). Conditioning on the post-treatment event D_{k-1} creates a collider path between treatment A and unmeasured frailty U (e.g., latent price sensitivity). As high-risk subscribers churn earlier in the control arm, surviving control subscribers become healthier, creating artificial crossover or attenuated hazard ratios over time.

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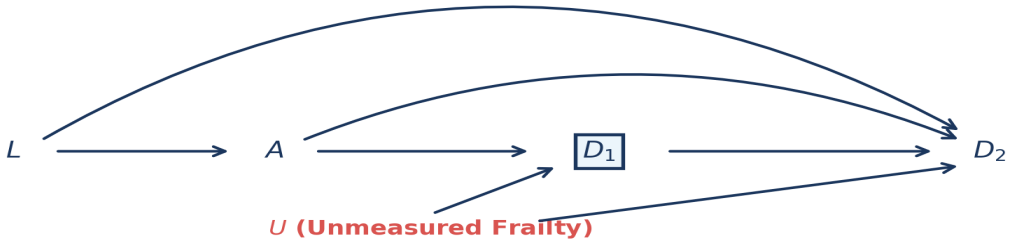


Figure 17.1: Selection Bias in Hazard Ratios (Conditioning on Survivor Collider D_1)

3. Longitudinal Person-Time Data Architecture

To enable causal survival analysis via g-methods, the subscriber database is converted from a wide format (1 row per subscriber) into a person-time long format (1 row per subscriber-month k). For a cohort of 10,000 subscribers followed over 120 months, this expands into up to 1.2 million person-month rows. At each month k, the binary event indicator D_k records whether churn occurred ($D_k = 1$) or the account remained active ($D_k = 0$).

Mathematical Foundation of Causal Survival Curves

Key Identity Connecting Hazards to Counterfactual Survival:

The counterfactual survival probability at month k equals the cumulative product of conditional survival probabilities:

$$P(D_k^a = 0) = \prod_{m=1}^{k} [1 - P(D_m = 1 \mid D_{m-1} = 0, A = a, L = l)]$$

By estimating discrete-time conditional hazards using pooled logistic regression, we can reconstruct full survival and risk curves.

4. Inverse Probability Weighting of Marginal Structural Survival Models

When baseline confounders L (e.g., historical tenure, monthly charges, support tickets) affect both intervention assignment A and time-to-churn T, unadjusted survival curves are confounded. **Inverse Probability (IP) Weighting** constructs a pseudo-population in which treatment A is independent of L.

- **Step 1 — Estimate Baseline Treatment Weights:** Fit a logistic model for $P(A = 1 \mid L)$ and compute stabilized weights $SW^A = \frac{P(A)}{P(A \mid L)}$.
- **Step 2 — Estimate Time-Varying Censoring Weights (if non-administrative censoring exists):** Compute $SW^C_k = \prod_{m=1}^k \frac{P(C_m = 0 \mid C_{m-1} = 0, A)}{P(C_m = 0 \mid C_{m-1} = 0, A, L)}$ to adjust for loss-to-follow-up.
- **Step 3 — Fit Weighted Pooled Logistic Hazards Model:** On the long person-time dataset, fit the model:

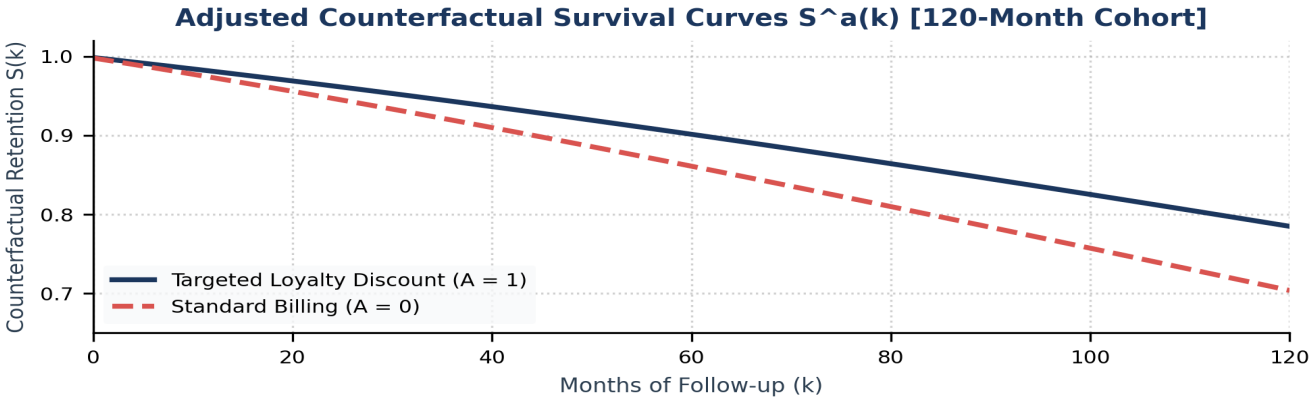
  $$\text{logit } P(D_{k} = 1 \mid D_{k-1} = 0, A) = \theta_{0,k} + \theta_1 A + \theta_2 (A \times k) + \theta_3 (A \times k^2)$$

  where $\theta_{0,k}$ is a flexible time baseline (e.g., cubic spline of k). Each subscriber-month row is weighted by $W = SW^A \times SW^C_k$.
- **Step 4 — Reconstruct Counterfactual Survival Curves:** Multiply $(1 - \hat{h}_a(m))$ across all $m \le k$ for $a = 1$ and $a = 0$.

5. Parametric G-Formula for Survival Outcomes

An alternative g-method for estimating marginal counterfactual survival $P(D_k^a = 0)$ is the **Parametric G-Formula** (standardization over the empirical covariate distribution $P(L)$):

- **Step 1 — Outcome Hazard Modeling:** Fit an unweighted pooled logistic regression for $P(D_k = 1 | D_{k-1} = 0, A, L)$ incorporating treatment, time splines, baseline confounders L, and $A \times L$ interaction terms.
- **Step 2 — Individual Counterfactual Prediction:** For each subscriber in the baseline dataset, predict monthly conditional hazards $\hat{h}(m | A=a, L_i)$ under $a=1$ and $a=0$ for all $m = 1 \dots 120$. Compute individual counterfactual survival $S_i^a(k) = \prod_{m=1}^k (1 - \hat{h}(m | A=a, L_i))$.
- **Step 3 — Empirical Standardization:** Average predicted survival curves across the population: $S^a(k) = (1 / N) \sum_{i=1}^N S_i^a(k)$



6. Structural Accelerated Failure Time (AFT) Models & Artificial Censoring

While IPW and the G-Formula estimate survival probabilities $P(T^a > k)$, **Structural Accelerated Failure Time (AFT) Models** directly model the ratio of counterfactual survival times: $T^a / T^{a=0} = \exp(-\psi a)$. The parameter $\exp(-\psi)$ represents the factor by which intervention expands median subscriber tenure.

> **G-Estimation of Structural AFT Models**
> **Artificial Censoring in G-Estimation:**
> Because administrative censoring occurs at cutoff $K = 120$, subscribers with $T_i > K$ have unknown failure times. Simply restricting g-estimation to uncensored subscribers introduces severe selection bias because treatment shifts survival times across the cutoff K. To preserve exchangeability, g-estimation uses **artificial censoring** $\Delta(\psi)$, artificially censoring subscribers whose counterfactual survival under alternative treatment would exceed $K(\psi) = K \exp(\psi A)$.

7. Benchmark Results & Decision Matrix for Production Pipelines

Survival Method	120-Month Risk $R^1(120)$	120-Month Risk $R^0(120)$	Causal Risk Difference	Primary Methodological Advantage
Naive Unadjusted	23.8%	18.0%	+5.8% (Confounded)	Simple, but severely biased by baseline age/tenure.
Unadjusted Cox HR	HR = 1.34	1.00 (Ref)	N/A (Hazard Ratio)	Non-collapsible; built-in selection bias over time.
IPW Marginal Structural	19.3%	23.5%	-4.2% (-6.1%, -2.3%)	Retains full sample; correctly adjusts for confounding.
Parametric G-Formula	19.1%	23.6%	-4.5% (-6.3%, -2.6%)	High precision; robust when outcome hazard is well-modeled.
Structural AFT (G-Est)	Median Tenure Ratio: $\exp(-\psi) = 1.22$		22% Tenure Expansion	Directly models tenure expansion; robust to censoring.

Production Pipeline Recommendation: In commercial telco retention systems, naive models suggest retention discounts increase long-term churn (+5.8% risk difference) due to adverse selection (discounts given to high-risk accounts). Causal survival g-methods (IPW and Parametric G-Formula) remove baseline confounding, revealing that retention discounts actually decrease 120-month churn risk by **-4.2% to -4.5%**. Production pipelines should privilege counterfactual survival differences $S^1(k) - S^0(k)$ over hazard ratios.